

AI-Powered Multi-Disease Risk Prediction System

Final Updated Project Explanation & Documentation

1. Project Overview

This project is a Machine Learning based Healthcare Analytics System developed using Python, Scikit-learn, Pandas, Streamlit, and data visualization libraries. The system predicts disease risk using patient medical data. Initially, the project focused only on Diabetes Prediction, but later it was upgraded into a Multi-Disease Prediction System by adding Heart Disease Prediction.

2. Final Project Features

- Diabetes Risk Prediction
- Heart Disease Prediction
- Streamlit-based Interactive Web Application
- Machine Learning Prediction Models
- Logistic Regression & Random Forest Models
- Confusion Matrix & Classification Report
- Health Analytics Dashboard
- Real-time Risk Prediction
- GitHub Integration
- Online Deployment Support

3. Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Streamlit
- Joblib
- Git & GitHub

4. Project Folder Structure

```
AI_Multi_Disease_Risk_Prediction_System/  
■■■ dashboard/app.py  
■■■ data/diabetes.csv  
■■■ data/heart.csv  
■■■ models/diabetes_model.pkl  
■■■ models/heart_model.pkl  
■■■ notebooks/eda.ipynb  
■■■ notebooks/heart_disease_model.ipynb
```

■■■ reports/
■■■ README.md
■■■ requirements.txt

5. Diabetes Prediction Workflow

The diabetes prediction system uses the diabetes dataset containing patient health measurements such as Glucose, BMI, Insulin, Blood Pressure, Pregnancies, and Age. Logistic Regression and Random Forest models were trained and evaluated. Logistic Regression was selected as the final model.

6. Heart Disease Prediction Workflow

The heart disease prediction system uses medical parameters such as age, cholesterol, blood pressure, chest pain type, and heart rate. Logistic Regression achieved approximately 79% accuracy for prediction.

7. Machine Learning Pipeline

1. Data Collection
2. Data Preprocessing
3. Exploratory Data Analysis (EDA)
4. Data Visualization
5. Feature Selection
6. Train-Test Split
7. Model Training
8. Prediction
9. Model Evaluation
10. Model Saving (.pkl)
11. Streamlit Web App Integration

8. Evaluation Metrics

The project uses the following evaluation techniques:

- Accuracy Score
- Confusion Matrix
- Classification Report
- Precision
- Recall
- F1-Score

9. Final Model Accuracy

- Diabetes Prediction Accuracy: ~74.67%
- Heart Disease Prediction Accuracy: ~79%

10. Streamlit Dashboard

The Streamlit application contains:

- Disease Selection Dropdown
- Interactive Input Forms
- Risk Prediction Button
- Analytics Dashboard
- Charts and Visualizations
- Dataset Preview Tables
- Health Recommendations

11. GitHub & Deployment

The project is uploaded to GitHub with proper documentation and project structure. The Streamlit application is deployment-ready and can be hosted online using Streamlit Cloud.

12. Project Conclusion

This project successfully demonstrates the implementation of Machine Learning in healthcare analytics. The system predicts disease risk using patient data and provides real-time prediction through a web application interface. The project evolved from a beginner-level diabetes notebook into a professional multi-disease healthcare analytics system.

13. Future Scope

- Add more disease prediction modules
- Improve accuracy using advanced ML algorithms
- Add patient login and history tracking
- Deploy application publicly online
- Integrate deep learning models
- Add real-time healthcare monitoring